

API endpoints summary

1. Keeping Things Safe & Simple

Farmers are busy people. They need an app they can trust without jumping through hoops to use it.

- **Getting Started:** Joining the platform is as easy as typing in a username, password, and email. To keep out spam and fake accounts, the app shoots a quick verification link to their inbox to make sure they are a real person.
- **Safe & Sound:** Once a farmer logs in, the app gives them an invisible, secure "badge" that keeps them logged in while they work.
- **Worry-Free Logouts:** If a farmer is using a shared computer (like at a public library or a shared office), hitting "Log Out" permanently destroys that digital badge. Nobody else can sit down at that computer and snoop through their farm data.
- **The Overseer's View:** For the people running the app (the admins), there is a private, organized list of everyone who has joined the community so they can keep things running smoothly.

It's incredibly easy to use, but built like a vault. Farmers never have to stress about their private data falling into the wrong hands.

2. The Digital Control Room

Imagine being able to check on your entire farm while sitting at your kitchen table having your morning coffee. That's what this does.

- **The Big Picture:** There's a public directory where anyone can browse a high-level list of all the farms connected to the network.
- **Claiming Your Land:** A farmer can register their own fields in seconds. The app instantly recognizes them as the rightful owner, meaning nobody else can mess with their data.
- **Making Changes:** Did you switch from growing wheat to potatoes? Did you move? Farmers can quickly update their farm's details anytime. (And if a farm shuts down, an admin can cleanly wipe it from the system).

- **Plugging in the Tech:** If a farmer buys a smart soil monitor or a temperature gauge, they can easily link that physical device right to their digital farm profile.
- **Hyper-Local Weather:** Forget watching the local news. With one click, the app pulls down the exact, real-time weather conditions for the farm's precise GPS coordinates.

It turns a physical piece of land into a smart dashboard. Farmers spend less time driving out to check fields and more time actually growing their business.

3. The Smart Assistant & Safety Net

This is where the app goes from being a digital notepad to an active, helpful farmhand that looks out for you.

- **The "Do I Water?" Feature:** This is a lifesaver. Instead of guessing, a farmer can check the app. It instantly reads their soil moisture sensors and gives a clear Yes or No on whether they need to turn on the irrigation that day.
- **Spotting Trends:** The app crunches all the past weather data to show farmers their true average temperatures and wind speeds over time, helping them spot seasonal changes on their specific plot of land.
- **Neighbor vs. Neighbor:** A farmer can look up their whole county (like Belfast or Derry) to see the average weather for the community. It helps them figure out if their specific farm is running hotter, colder, or drier than their neighbors.
- **Easy Searching:** Looking for a specific farm? Just type in the town name or the postcode and hit search.
- **Emergency Alarms:** If a freak frost, wildfire, or flood is coming, an admin can draw a "danger zone" on a digital map. Every single farmer inside that specific area will instantly get an emergency warning.

It's not just holding data; it's actively protecting the farmer. It saves water, protects crops from sudden disasters, and takes the total guesswork out of daily chores.

Meth od	Endpoint	Descriptio n	Authenticatio n Required	Response
POST	/api/users/register	Creates a new user account and generates a verification link.	NO	Req: {"username":"x", "password":"y", "email":"z"} Res: {"message": "Account created...", "verification_link": "..."}
GET	/api/users/verify/<user_id>	Verifies a user's email address by flipping is_verified to True.	NO	Req: None Res: {"message": "Email successfully verified!"}
POST	/api/login	Authenticates a user and returns a 30-minute JWT token.	Basic Auth (Username/Pass)	Req: Basic Auth Headers Res: {"token": "eyJhb...", "role": "user"}
GET	/api/logout	Adds the user's current JWT to a database blacklist to terminate the session.	Yes (JWT)	Req: Bearer Token Res: {"message": "Logout successful"}
GET	/api/users	Retrieves a list of all registered users	Yes (Admin JWT)	Req: Bearer Token

		(excluding passwords).		Res: {"count": 3, "users": [...]}
GET	/api/farms	Retrieves a summary list of all registered farms.	NO	Req: None Res: [{"farm_name": "Plot 1", "crop_type": "Wheat", ...}]
GET	/api/farms/<farm_id>	Retrieves the full document (including sensors/logs) of a specific farm.	NO	Req: None Res: {"_id": "...", "sensors": [...], "weather_logs": [...]}
POST	/api/farms	Registers a new farm and links the owner_id to the current user.	YES(JWT)	Req: {"farm_name": "My Farm", "crop_type": "Wheat"} Res: {"farm_id": "...", "message": "Registered!"}
PUT	/api/farms/<farm_id>	Updates safe top-level farm details (e.g., crop type, location).	Yes (Owner/Admin JWT)	Req: {"crop_type": "Barley"} Res: {"message": "Farm updated successfully!"}
DELETE	/api/farms/<farm_id>	Permanently deletes a farm record	Yes (Admin JWT)	Req: Bearer Token

		from the database.		Res: {"message": "Farm deleted successfully"}
POST	/api/farms/<farm_id>/sensors	Pushes a new IoT sensor object into the farm's sensors array.	Yes (Owner/Admin JWT)	Req: {"sensor_id": "S1", "type": "Soil Moisture"} Res: {"message": "Sensor added", "sensor": {...}}
GET	/api/farms/search?q=<term>	Uses compound text indexing to search farm area names and postcodes .	NO	Req: ?q=Belfast Res: {"results_count": 2, "data": [...]}
POST	/api/farms/<farm_id>/sync_weather	Calls the external Open-Meteo API using farm coordinates to fetch live weather.	Yes (Owner/Admin JWT)	Req: None Res: {"message": "Weather synced", "new_log": {...}}
POST	/api/farms/alerts/broadcast	Uses \$geoWithin to find farms in a polygon and push emergency alerts.	Yes (Admin JWT)	Req: {"alert_type": "Flood", "danger_zone": {...}} Res: {"farms_notified": 5, "alert_type": "Flood"}

GET	/api/farms/<farm_id>/insights	Uses \$unwind and \$group to calculate a farm's average temp and windspeed.	Yes (Owner/Admin JWT)	Req: None Res: {"dashboard_data": {"average_temp": 12.5, ...}}
GET	/api/farms/<farm_id>/irrigation_check	Reads the latest Soil Moisture sensor value to advise on watering	Yes (Owner/Admin JWT)	Req: None Res: {"status": "WARNING", "moisture_level": 15.5}
GET	/api/farms/region/<region>/insights	Aggregates logs across multiple farms in a region to find community averages.	NO	Req: None Res: {"data": {"community_avg_temp": 14.2, ...}}